
TWO NAVIER–STOKES PAPERS, ONE INSTRUMENT

*A comparative audit supplement for the general reader: a verified foothold set
beside a claimed summit, with falsifiability as the deciding axis*

AUDIT SUPPLEMENT

*To state, in one table a non-specialist can read, what each paper claims, what each has earned, and which of
the two can be tested against the world*

TRISDUCTION AUDIT SERIES · SUPPLEMENT TO DOI 10.5281/ZENODO.22665831

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8 September 2026 · sealed hardened edition v2, audit cycles supp-20260908 (v4) and supp2-20260908 (v2)

ABSTRACT. Two papers on the three-dimensional Navier–Stokes equations were read by the same instrument on the same day. The first, Integrable Misalignment Forbids Blowup, is an eight-page paper that proves two continuation criteria from the classical base and closes the regularity question only under one disclosed premise. The second, Finite Time Blowup for Navier–Stokes, published under the author line OPENAI, is a 166-page construction claiming a smooth force that drives a viscous fluid to infinite velocity in finite time at bounded energy, which if correct resolves two of the four Millennium alternatives. This supplement compares them on identity, method, proof surface, verification state, hidden assumptions, parsimony, induction, falsifiability, physical anchoring, and provenance, and states three rulings. Neither paper is certified: both sit at the same seat, awaiting a verifier who is not the author. The axis that separates them is falsifiability. The first paper names in advance what would refute it and can be met by a fluid or a computer; the second, by its own exponents, places its singularity beyond every finite observation window, so nothing in the world can reach it and only a referee can. The second paper is also, if its towers close and to this auditor’s knowledge, the first hard witness of a Navier–Stokes blowup in circulation, and the first paper’s falsifier survives contact with it: extended to a smooth force, the criterion predicts that any such blowup carries a divergent misalignment integral, and a scaling of the construction’s core gives exactly that divergence. The second paper also ships a Lean 4 formalization bound to a third-party formal statement of the Clay problem, which moves its certification question from prose to a build any reader can run; that build has not yet been reproduced by anyone outside the claimant, and a concurrent human-and-model program by Buckmaster and Alpöge, with Lean-verified forced blowups for Euler and two related equations and a public statement raising timing and data questions, holds the provenance door open. The unforced existence-and-smoothness problem, the one most people mean by the Millennium problem, remains open, and only the first paper addresses it. The supplement closes with a census of every infinity each paper asserts, under a strict placement law, and finds that the first paper asserts none.

KEYWORDS *Navier–Stokes; regularity; finite-time blowup; audit; falsifiability; parsimony; provenance; Millennium problem*

1 HOW TO READ THIS SUPPLEMENT

The two papers are not rivals. One builds a verified foothold on the unforced equation; the other attempts the summit of the forced one. An auditor cannot rank a foothold against a summit by ambition, so the rulings below rank them by two things a reader can check: how much of each claim is on the page at a stated grade, and whether anything in the world could ever contradict it. Every executed check in this supplement was run in the open, and the numbers quoted are the ones the runs produced. Where the audit could not verify a step, the table says so; an unverified step is not a broken one, and the word “held open” marks exactly that.

Three words carry weight here. A paper is *deductive* when its conclusions follow from stated inputs by finite steps that a reader can check one by one. A paper carries *induction* when it builds an object in infinitely many stages and asserts the limit exists, which is where most long constructions fail. A paper is *falsifiable* when its author names, in advance, an observation or a computation that would prove it wrong. Three further words are borrowed from the audit instrument. A claim is *sealed* when the audit has verified it at a stated grade. A *gate* is one of twelve structural checks a claim must pass, and a gate is *held open* when the audit could neither

pass nor fail it. The *seat* is a paper's certification state: who has verified it, and whether that verifier is independent of its author. Four more words belong to machine-checked mathematics. A *formalization* is a proof rewritten in a language a computer can check line by line; the *kernel* is the small trusted core of that checker; a *build* is the run in which the kernel checks every line; a *reference statement* is an independently written formal version of the problem, so that a proof can be compared against the problem as posed rather than as the prover restated it, and *permitted axioms* are the few standard assumptions a checked proof is allowed to rest on.

2 THE TWO PAPERS IN ONE PARAGRAPH EACH

Integrable Misalignment Forbids Blowup (IMFB) introduces a single new quantity, the alignment defect, which measures how far neighbouring vortex directions disagree, weighted by their strengths and their distance. It proves that if a time integral of this quantity stays finite, the flow cannot blow up, first for a family of subcritical members and then at a scale-invariant endpoint. It then states one premise, that the defect admits a data-uniform bound, and shows that this premise alone closes global regularity. The premise is labelled as a premise. The paper was audited in three adversarial cycles before sealing; the first cycle found the original lemma wrong and forced its repair.

Finite Time Blowup for Navier–Stokes (FTB) prescribes a flow, a spinning column that contracts to a point at time one, and defines the external force to be whatever the equation needs, then spends some 160 pages proving that this force stays smooth while the velocity diverges. The construction sums corrections in two infinite towers, uses oscillatory pulses to supply the missing stress, and tunes more than a dozen parameters in a strict order. If every estimate holds, it resolves Millennium alternatives (C) and (D), the forced variants; the unforced problem, alternatives (A) and (B), is untouched by it. The author line is OPENAI and no human is named. OpenAI's announcement of 8 September 2026 describes the process: an internal model beyond GPT-6 Astra, on the order of ten thousand coordinating agents, 88 hours, 2.7 million messages, some 130 billion output tokens, started on 1 September after rumors of a rival result, with the unforced Euler blowup found first as a warm-up. It ships a Lean 4 formalization whose reference statement is adapted from an independent third-party formalization of the Clay problem, with only the three standard axioms permitted and instructions for anyone to re-check it. No peer review exists, the Clay Institute has verified nothing, and OpenAI states it does not intend to claim the prize. Concurrently, Tristan Buckmaster and Levent Alpöge released Lean-verified finite-time blowups with smooth forcing for the porous-media, Boussinesq, and three-dimensional Euler equations, built on the program of Córdoba and Martínez-Zoroa with model assistance, with a hypo-dissipative Navier–Stokes result withheld pending its own formal check; Buckmaster's public statement raises questions of timing, data access, and authorship pressure that OpenAI's announcement answers in part and denies in part. The audited copy of the OpenAI paper cites the Córdoba and Martínez-Zoroa program; the public copy fetched the same day does not carry that paragraph.

3 THE MASTER TABLE

TABLE 1 | THE MASTER COMPARATIVE TABLE, GENERAL-AUDIENCE EDITION. IMFB IS THE SEALED ALIGNMENT-DEFECT PAPER; FTB IS THE OPENAI BLOWUP CONSTRUCTION. RULINGS ARE THE AUDITOR'S, TYPED AT THE GRADE EACH ROW EARNS.

Axis	IMFB, alignment-defect paper	FTB, OpenAI blowup construction	Ruling
What it is	8 pages; two unconditional criteria plus one conditional closure	166 pages; an explicit blowup under a smooth force	A foothold and a summit attempt
Equation	Unforced Navier–Stokes	Forced Navier–Stokes	Different problems; no collision
Claim strength	Modest by design	Two Millennium alternatives resolved, if correct	FTB maximal
Which Clay statement	The unforced problem, alternatives (A) and (B), the one most people mean, approached through a criterion	The forced variants (C) and (D)	The unforced problem remains open for everyone; only IMFB is about it
Authorship and provenance	Human-seeded, AI-assisted; written, read three times in adversarial audit, and sealed with the record published	An AI agent system with no named human author; the process self-reported by the claimant; no referee record; a concurrent human-and-model program on the same route with a public dispute over timing, data, and credit	One paper shows its reading; the other shows its process and its dispute
Where it lives	Zenodo DOI 10.5281/zenodo.22665831, Internet Archive, GitHub, content-hashed	An OpenAI announcement dated 8 September 2026, a CDN link with no DOI, and a public GitHub Lean repository; the public copy and the audited copy differ in their reference list, so at least two versions circulate	One is on a permanent DOI record; the other's formal artifact is public and pinned to a toolchain
Deduction	Finite chains from five cited theorems; every step on the page	Deductive in form; the core is two infinite towers completed by a limit	IMFB pure deduction; FTB deduction resting on induction
Induction	Zero inductive steps	Two towers: a background expansion and a correction cycle, both summed to infinity	FTB carries all the inductive risk
Parsimony of assumptions (Occam's first blade)	Five classical inputs plus one disclosed premise	No premise beyond the classical base; it constructs rather than assumes	FTB assumes less
Parsimony of machinery (Occam's second blade)	One free scale, printed constants, eight pages	A chain of more than a dozen scheduled parameters, two towers, a positivity cone, an auxiliary torus	IMFB spends less; the razor cuts differently on each blade
The small number h	Not present	Displayed as $h < 1/100$, actually fixed last in a chain that makes it smaller than	The construction's currency of progress is a tower-

Axis	IMFB, alignment-defect paper	FTB, OpenAI blowup construction	Ruling
		$\exp(-\exp(M_d))$, a tower of exponentials; every gain in the construction is measured in powers of q^h	exponentially small power of time
Falsifiability, stated in advance	Two falsifiers named: a smooth-data blowup with finite defect integral, or a derivation of the premise from the bare regularity question	None named; refuted only by finding a gap in the proof	IMFB is the falsifiable paper
Falsifiability, reachable in the world	A fluid simulation or a laboratory flow can meet the criterion and its falsifier	By the paper's own exponents the critical norm diverges as $\tau^{-4h/3}$, invisible at every finite time before the singularity; no experiment, no simulation, no finite reader can meet it	FTB is free-floating for a real-world problem
Physical anchoring	Vortex alignment and its depletion are measured turbulence facts; the criterion reads a geometry that exists	The singularity forms below every physical scale under a force acting at ordinary scales; no actual fluid follows it	IMFB anchored, FTB an idealization it does not name as one
Proof surface and blast radius	One error breaks one lemma	One wrong inequality breaks the theorem	Small against total
Executed consistency checks	Every exponent re-executed; endpoint scale invariance confirmed	Every leading scaling re-executed by the auditor and consistent with every known necessity theorem: energy, critical norm, dissipation, vorticity, viscosity map	Both pass every scaling check they must
The one place a referee should look	Whether the criterion is already implied by a known direction-regularity result	Whether the Lean statement is the Clay statement and whether the build reproduces on an independent machine; the prose seams of Lemmas 10.2 and 10.3 become moot if it does	A literature check against a build and a statement-match audit
Hidden assumptions and disclosure	Premise disclosed; ranges disclosed; roster-relative novelty flagged	True size of h behind a cap; no constants table; no statement of physical scope; abstract reads unforced until the theorem; the formal build, if it reproduces, discharges the constants a hand check cannot reach	Neither launders a premise; FTB's hand-unreachable content is machine-reachable
Consistency with each other	Extended to a smooth force, its criterion predicts a	A crude scaling of its core gives exactly that divergence	

Axis	IMFB, alignment-defect paper	FTB, OpenAI blowup construction	Ruling
	divergent defect integral for any such blowup		If FTB is right it corroborates IMFB; IMFB says nothing about FTB
First hard witness	Its stated falsifier survives contact with the first hard blowup witness known to this auditor, and the witness corroborates the criterion at scaling grade	Is that witness, if its towers close	Corroboration, not proof; typed, load-bearing on nothing
Verification state	Three audit cycles, twenty findings dispositioned and fifteen earned, five auditor errata in the second cycle and none in the first or third; no independent verifier yet	Kernel-checked by the claimant against a third-party reference statement with standard axioms only; no independent reproduction on record; one public objection standing on provenance	Both uncertified; one has been read by a human, the other by a kernel
Closure path	An independent referee, a machine-checked proof, or a computational reproduction; the third partly walked, the second specified but not built	The mechanical road walked by the claimant; an independent re-run of the build and a statement-match audit outstanding; the human road not begun	FTB is ahead on the mechanical road; neither is sealed
Superior as verified knowledge today	Yes		Everything on the page at a stated grade
Superior in consequence if certified		Yes, and not on the same scale	A Millennium alternative
More reality-anchored, less likely to be falsified	Yes		The price is that it says less

Box 1 | WHY FALSIFIABILITY DECIDES THE REALITY QUESTION

A paper about a real-world problem earns its anchor by exposing itself to the world. IMFB does this twice: it names two falsifiers before anyone asks, and both are reachable, since a numerical flow that blew up while keeping the defect integral finite would refute it on a screen. FTB exposes itself to nothing but referees. The reason is arithmetic rather than rhetoric. Its blowup rate is $\tau^{-1/2-h}$ with h fixed at the bottom of a parameter chain that makes it smaller than $\exp(-\exp(M_d))$; the quantity that must diverge for any singularity, the critical norm, diverges as $\tau^{-4h/3}$, which is indistinguishable from constant at every time a computer or a fluid can reach. The construction is therefore unobservable by design. That is not a logical flaw, and this supplement finds none. It is the precise meaning of free-floating: a claim about a real-world problem that no part of the world can ever touch, whose only judge is a reader who is not its author.

4 THE FALSIFIERS AND THE FIRST HARD WITNESS

IMFB states two falsifiers in its published Zenodo version, dated 8 September 2026, and the audit opened the OpenAI file after that record existed. F1: exhibit one smooth-data solution of the unforced equation that blows up in finite time while its integrated alignment defect stays finite. F2: derive the paper's one premise from the bare regularity question by base means, which would show the premise is the problem in disguise. Either event sinks the paper, and both are stated so that a reader can go looking.

The OpenAI construction, if its towers close, is, to this auditor's knowledge, the first hard witness of a Navier–Stokes blowup in circulation: not an averaged model, not a weak solution, an explicit smooth flow under a smooth force with infinite velocity at time one. It is forced, so as stated it cannot meet F1, which is unforced. But the criterion extends to a smooth compactly supported force by one line, not in the published paper and sketched in the audit record: the enstrophy balance gains a term bounded by the L^2 norm of the curl of the force times the L^2 norm of vorticity, its coefficient integrable in time, so the continuation theorems survive and predict that any forced blowup carries a divergent integrated defect. A crude scaling of the construction's core, direction varying across a radial width $q^{1/2}$ with vorticity of order τ^{-1-h} , gives the defect integral an exponent of $-1 - 4h$, divergent. The theorems and the two falsifiers were on record before the witness was read; the forced extension was written in the audit after reading it, and its prediction was then checked against the construction's scaling. The witness does not refute the criterion, and it agrees with the extension at the level of exponents.

This is indirect corroboration, reached after the fact and typed exactly so: corroboration grade, load-bearing on nothing, because Theorems 4 and 6 are proved and need no vote. What it shows is what a falsifiable claim looks like when the first hard object arrives: the door stayed open, something walked through, and the claim stood. A criterion that had said nothing about forced flows could not have been touched; this one could, and was not.

5 THE GEOMETRIC GROUND, WHY IMFB STANDS

The object is a shape. The alignment defect reads the direction field of vorticity and nothing else: it vanishes exactly where neighbouring directions agree, it is unchanged by rotating or reflecting the frame, and at each member's endpoint it is unchanged by the equation's own scaling, exponent zero, executed. Quantities invariant under the symmetries of their setting are what count as structure; distances, addresses, and magnitudes of gaps are artifacts of the chart and the defect carries none of them. The mechanism is not a fit: the depletion of vortex stretching where directions align is cited as an input, and the functional is built to match the stretching term exactly, so that the enstrophy power cancels to two for every member of the family as an algebraic identity. The seal sits on the geometry and the truth-value stays open, the one premise printed as a premise. And it lives where physics lives: alignment and depletion are measured in turbulence, and the criterion can be evaluated on any simulation.

6 THE INFINITY CENSUS, STRICT PLACEMENT

The placement law is stated once. A completed infinite tower is groundless unless a uniform bound anchors it; importing a tower's limit as if it were a value is the Cantorian artifact trap; a finite object whose existence proof

needs infinitary strength carries that strength as its price. Every infinity each paper asserts is then listed and typed.

TABLE 2 | THE INFINITY CENSUS. EACH ROW IS ONE INFINITY ASSERTED OR USED BY ONE PAPER, WITH ITS KIND, ITS ANCHOR, AND ITS PLACEMENT UNDER THE STRICT LAW.

Infinity	Kind	Anchor	Placement
FTB: unbounded velocity at $t = 1$	Actual infinity, the theorem's object	Exists only if every other row closes	A constructed object if the towers close; no actual fluid carries an infinite rate
FTB: background expansion in q^{2h}	Completed tower	Borel cutoffs with radii $\log 10$ of the cutoff radius of order $0.9/h$	Anchored in form, open in content
FTB: correction cycle in $\varepsilon = q^h$	Completed tower	Borel tails 2^{-j}	Anchored in form, open in content
FTB: pulse frequencies $\rightarrow \infty$, separation $q^{h/2}$	Potential infinity	Viscous damping balanced at order one	Legitimate; the separation vanishes at every finite time
FTB: the parameter h below $\exp(-\exp(M_d))$	Finite but tower-exponential	Fixed in order, acyclic at every sentence read	Finite in law, infinite to every observer; the price of the object
FTB: the singular time $t = 1$	Limit point of the construction	The derivative limits of Lemma 10.2	The one door, held open by this audit
IMFB: $T^* = \infty$, global existence	Potential infinity	None needed	The conclusion of a forbidding, not an exhibiting
IMFB: the defect integral and the Lorentz norms	Classical integrals	Cited theory	Standard analysis, no tower

Reading. FTB asserts one actual infinity and reaches it through two completed towers whose form is anchored and whose content is unverified, driven by a parameter so small that every gain the construction makes is a power of q^h with h below a double exponential. The law does not call this false; it holds it at arm's length until a reader who is not the author closes the anchor. IMFB asserts no completed infinity. Its only infinity is the global time it forbids the flow from failing to reach, a potential infinity, and it is the one entry in the census that needs no anchor because it is the conclusion of a forbidding.

7 FOR AND AGAINST, TYPED

For IMFB, five pillars. Geometry over force: it names the intrinsic price of blowup, misalignment, whatever the force does. Zero induction: finite deductive chains, no tower, no limit asserted. Parsimony on machinery: one premise, eight pages, printed constants, one free scale. Falsifiable and reachable: two falsifiers named in advance, both meetable by a simulation, and the first hard witness did not meet them. Read before published: three adversarial cycles with the first error on the record. About the unforced problem: it is the one paper here written about the equation whose regularity the prize actually asks about.

Against FTB, five charges, each typed. Free-floating on a real-world problem: the critical norm diverges as $\tau^{-4h/3}$, which at the displayed cap $h = 1/100$ grows by twenty percent over six decades of time and at the working

value of h by nothing any instrument resolves; unobservable by design, executed. Tower-tuned: h is fixed last in a chain that puts it below $\exp(-\exp(M_d))$, the outer-tail exponent window has width h , and both towers advance by powers of q^h ; executed from its own schedule. Anchored in form only, to prose: two completed towers closed by Borel devices whose constants across tower-exponentially many stages no prose audit reaches, and the constants table is unprinted; held open in prose, dischargeable by the kernel if the public build reproduces. Human-unread: no named author, no referee, no independent reproduction of the build on record; the kernel has read it for the claimant, and a public objection on provenance stands unanswered on the record. Undisclosed scope: an idealization never named as one, and an abstract that reads unforced until the theorem.

The charge not made: that the mathematics is wrong. The audit found no broken gate, no contradicted theorem, no dangling reference, and every leading scaling consistent with every necessity theorem. Conceded in the other direction: FTB assumes nothing and, if certified, carries a weight IMFB cannot approach; IMFB says less and knows it; both seats stand uncertified; and if FTB is right it corroborates IMFB while IMFB says nothing about FTB. The two are not rivals. One is the ruler; the other is the exception the ruler was built to measure.

8 WHAT THE AUDIT FOUND ON THE OPENAI PAPER, IN PLAIN TERMS

The audit ran the paper's central claim through a linguistic parse, twelve structural gates, a re-execution of its scaling exponents, a certification-state check, and a targeted second pass on the four places most likely to hide a gap. Ten of the twelve gates pass. Two are held open at one address: the infinite tower where every derivative of the residual must converge while the flow diverges. No known theorem is contradicted. The scaling exponents, energy $1/2 - 3h$, critical norm $-4h$, dissipation $-1/2 - 3h$, vorticity $-1 - h$, all sit exactly where a genuine singularity would need them. The parameter schedule is one-directional at every sentence read; the constants table that would let a reader check it by hand is not printed. On the outer tail, the exponent the positivity cone admits lies in a window of width h . The seat, before the formalization was known, read: witness present, certification refused, because the verifier is the claimant. With the formalization read, it reads differently and more precisely. A Lean 4 project exists in public, pinned to Lean 4.34.0-rc2 and Mathlib, exposing two theorems named for alternatives (C) and (D) whose statements are adapted from an independent third-party formalization of the Clay problem, with a checking configuration that permits only the three standard axioms and instructions for anyone to run the comparison. This audit did not run the build, which takes hours on a full Mathlib cache, and records the artifact as read. If an independent machine reproduces it and an independent reader confirms the adapted statement is the Clay statement, the tower seams held open above are discharged by the kernel, which does not care how small h is. Until then the seat reads: machine witness present, reproduction and statement match owed to a reader who is not the claimant.

Two corrections were made to the auditor's own findings under audit symmetry, and both are recorded. The first conflated the background expansion with the correction cycle. The second, forced by the public copy of the paper, found that the correction cycle's gain of one tenth per stage is measured in the normalized unit $\varepsilon = q^h$, so that in physical time both towers advance by powers of q^h and both are h -scaled. The fragility signature is therefore total rather than local, and the earlier statement that the correction cycle was "the healthiest part" is withdrawn.

9 THE CONCURRENT WORK, THE DISPUTE, AND WHERE IT SITS

On the same day, Buckmaster and Alpöge released three finite-time blowups with smooth forcing, for the incompressible porous-media equation, Boussinesq, and three-dimensional Euler, each Lean-verified, and stated that a hypo-dissipative Navier–Stokes blowup exists but is withheld until its own formal check finishes. They credit the route, forced blowup through a smooth force, to the program of Córdoba and Martínez-Zoroa, whom they took as their starting point and pushed to smooth forcing with heavy model assistance. Buckmaster's statement reports that OpenAI's effort began after word of their progress circulated, that the initial description of the OpenAI result as a lone model with little human input did not survive the first call, that questions about training on their sessions went unanswered, and that publication and authorship arrangements were proposed which he declined. OpenAI's announcement states that its effort began on 1 September on hearing a rumor, that it recognizes Buckmaster and Alpöge's priority on forced Euler, that neither its researchers nor its agents saw their work, that no user data was accessed, and that it cannot rule out de-identified usage data having improved its models.

Where this sits in the instrument is exact and it is not where the headlines put it. None of it touches the mathematics: a theorem is true or false regardless of who was prompted when, and consensus, credit, and motive carry zero weight in either direction here. All of it touches provenance, which is a register of its own: an unretracted public objection on timing, data, and authorship is exactly the kind of standing item that holds the certification door open until it is answered on the record, and the door is held open on that ground and no other. The lineage is now disclosed in the audited copy, which cites Córdoba and Martínez-Zoroa; that the public copy fetched the same day lacks the paragraph is recorded, not judged.

One consequence for the comparison is plain. Every result released this week, OpenAI's and Buckmaster and Alpöge's alike, concerns a forced fluid equation or an inviscid one. The unforced viscous problem that carries the prize as most people understand it remains open, and IMFB is the one paper in this supplement written about that problem.

10 RULINGS

Superior as verified knowledge today: IMFB, because its every claim is on the page at a stated grade, its blast radius is one lemma, and it carries a published record of its own correction. Superior in consequence if certified: FTB, without contest and not on the same scale. More reality-anchored and less likely to be falsified: IMFB, on both counts, and the price is that it says less. The asymmetry that closes the comparison: FTB being right costs IMFB nothing and corroborates its criterion at the level of exponents, while IMFB being right says nothing about FTB. Both seats move only when a verifier who is not the claimant reads the proof or reproduces the build, and no byline, page count, press cycle, dispute, or consensus moves either.

11 AUTHOR'S PROVENANCE AND METHOD DISCLOSURE

IMFB was seeded by the author, drafted with an AI scribe under the author's direction, and read three times in adversarial audit cycles whose records are published beside the paper; the entry version's lemma was found wrong in the first cycle and repaired. This supplement was produced the same way and audits the author's own

paper with no quarter given. The instrument is the Trisduction verification framework: a three-state verdict economy, warrant typing on every claim, zero evidential weight for consensus or authorship in either direction, and the rule that no self-check is a witness. Every figure quoted was computed in the session that produced this supplement and is reported as produced. The FTB copy audited is identified by its SHA-256 prefix 0e779481; the public CDN copy was fetched on 8 September 2026 and found to be a variant build with different reference numbering, which is recorded rather than resolved.

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